



Internship at Sakha Global

“Smart Contract Deployment through the web”

Submitted by:

Amogh K

PES2UG22CS061

Under the guidance of

Supreeth S K,
Principal Software Engineer ,
Sakha Global

From 24th June 2024 to 24th August 2024 (2 Months)

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FACULTY OF ENGINEERING
PES UNIVERSITY**

(Established under Karnataka Act No. 16 of 2013)
100ft Ring Road, Bengaluru – 560 085, Karnataka, India

July 1, 2024



Amogh K
PES University,
Bangalore

Dear Amogh,

Sub: **Internship Offer**

Congratulations!!

We are pleased to offer you an internship program with **Sakha** for a period of **2 months**. Please find the following confirmation of the specifics of your internship:

Position Title: Intern
Start Date: June 24, 2024
End Date : August 23, 2024
Stipend : Rs 5,000/Month
Mentor: Supreeth SK
Min Work Hours Per day : 8 hours

Should you have any questions regarding the specifics of your internship, please contact me by phone (888.456.9324) or by email (thejaswini.kulkarni@sakhaglobal.com).

We look forward for your joining and we appreciate your interest in **Sakha Global**.

Best of Luck!!

Thejaswini Kulkarni

Head – Human Relations

DECLARATION

We hereby declare that the project entitled “**Audit Block**” has been carried out at by me under the guidance of Supreeth S K, Principal Software engineer and submitted in partial fulfilment of the credits for the degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester June 2024 to July 2024. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

PES2UG22CS061

Amogh k

A handwritten signature in blue ink, reading "Amogh k", is displayed within a rectangular box. The signature is written in a cursive style with a horizontal line underneath the name.

ACKNOWLEDGEMENT

I would like to express my gratitude to my guide Supreeth S K, Principal Software engineer for their continuous guidance, assistance and encouragement throughout the development of this project.

I am grateful to the internship coordinators Prof. Animesh Giri, Dr. Prema R, Dept. of Computer Science and Engineering, PES University- Electronic City Campus for organizing, managing and helping out with the entire process.

I take this opportunity to thank Dr. Sandesh B J, Chairperson, Department of Computer Science and Engineering, PES University – Electronic City Campus, for all the knowledge and support I have received from the department.

I would like to thank Dr. B.K. Keshavan, Dean of Faculty, PES University for his help.

I am deeply grateful to Dr. M. R. Doreswamy, Chancellor, PES University, Prof. Jawahar Doreswamy, Pro Chancellor – PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University & Prof. Nagarjuna Sadineni, Pro- Vice-Chancellor, PES University for providing to me various opportunities and enlightenment every step of the way.

Finally, this internship could not have been completed without the continual support and encouragement I have received from my parents and my friends.

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ABSTRACT

1. Introduction

Sakha Global is a dynamic and innovative company committed to driving technological advancements and fostering sustainable development. With a strong focus on delivering high-quality solutions across various industries, Sakha Global has established itself as a leader in implementing cutting-edge technologies and providing comprehensive services that cater to the evolving needs of its clients. The company's dedication to excellence and innovation has positioned it at the forefront of global business, enabling it to create impactful and transformative solutions.

The internship program at Sakha Global was part of the company's strategy to enter into upcoming technologies like blockchain. The internship in Sakha Global's Audit Block offers a unique opportunity for aspiring professionals to gain hands-on experience in the intersection of blockchain technology and project management. Interns will be immersed in the development and implementation of the Project Audit Block, working alongside seasoned experts to understand the intricacies of blockchain-based auditing, this was done on the amoy testnet and connected to it from the backend which used node, express, ethers.js

. This program was structured to provide interns with a comprehensive understanding of the real-world applications of web development, specifically focusing on the creation of a responsive, user-friendly website using the MERN stack—MongoDB, Express.js, React.js, and Node.js. Spanning a duration of two months, the internship required participants to commit to approximately 30 hours of work per week.

The program was delivered in a hybrid format, blending both in-office and remote work experiences, with 80% of the work being conducted on-site and the remaining 20% completed remotely. This hybrid structure was designed to offer interns the flexibility of remote work while also providing them with the invaluable experience of collaborating in a professional office environment.

Additionally, interns were expected to actively participate in team meetings, contributing ideas and solutions to various challenges that arose during the project. This collaborative aspect of the internship was intended to simulate a real-world working environment, where teamwork and communication are essential to the success of any project

2. II. Job Responsibilities and Duties

In this internship, the primary responsibility was to collaborate closely with other employees on developing a proof of concept (PoC) for a blockchain web application, a strategic initiative for the company. Over the course of two months, contributing approximately 30 hours, the role involved working alongside blockchain experts and the development team to design and implement the project's architecture using Node.js, Express, React, SQL, and ethers.js. This collaboration involved regular meetings to discuss blockchain integration strategies, troubleshooting technical challenges, and refining the application's features to ensure seamless interaction with the Ethereum network. The internship emphasized teamwork, continuous learning environment knowledge-sharing, and leveraging the expertise of senior developers to create a functional prototype that showcased the potential of blockchain technology within the company's broader objectives.

Tasks Assigned and Project Details

1. The initial tasks of the internship focused on building a strong foundation in blockchain development. This began with setting up the development environment by installing essential tools like Node.js, Express, Truffle, and Nodemon. Following this, there was a deep dive into blockchain fundamentals, covering consensus mechanisms, cryptocurrencies like wei and gwei, gas prices, Web3, and decentralized finance (DeFi), along with the basics of APIs and HTTP methods like GET and POST. The learning process then shifted to Solidity, where key concepts such as data types, mappings, arrays, modifiers, and events were covered. This knowledge was then applied by writing smart contracts using the Remix IDE, providing a practical understanding of blockchain programming.
2. The second task involved developing APIs using Node.js and Express to handle GET and POST requests, with support for various data types like arrays, strings, and numbers. The focus was on creating APIs to interact with smart contracts using the ethers.js library, including contract creation, function calls, and parameter handling. This task provided hands-on experience in API development with the help of postman and smart contract management. And expanded on this by creating APIs to retrieve details of smart contract arguments and modifying the createEvent API to accept multiple arguments and events. These enhancements improved the flexibility and scalability of the smart contract interactions, reinforcing key concepts in API design and blockchain integration.
3. In this phase, the focus was on creating and integrating a database into the project. This involved establishing a connection to the database, saving details with a proper schema, and updating the APIs to fetch data from the database instead of local storage. The modifications improved data management and ensured the application was backed by a reliable database system(mysql).
4. In the next phase, the focus shifted to learning React and applying that knowledge to develop the frontend of the application. This involved creating a user-friendly landing page, a home page that featured a list of contracts, and dedicated pages for creating contracts, listing functions, and calling functions. This phase was crucial in building the user interface, ensuring smooth navigation and interaction within the web app. The pages of the webapp are mentioned below

Landing Page:

- The entry point of the web app, designed to provide a welcoming interface for users.
- Includes an overview of the application's purpose and functionalities.
- Features easy navigation to other key sections of the web app.
-

Home Page:

- Serves as the main dashboard, offering users a centralized view of their activities and relevant information.
- Shows all the smart contracts deployed on to the Ethereum network along with the name ,deployment date and navigation to the next page
- Summarizes the status of deployed contracts, recent transactions, and any updates.
- Provides links to navigate to the function list page and other key features.

Function List Page:

- Lists all available functions within the selected smart contracts.
- Each function is categorized and described for easy identification and use.
- Users can select specific functions to learn more or to proceed with interaction.

Call Function Page:

- Allows users to interact directly with the smart contract functions.
- Provides a form-based interface for entering required parameters and executing functions.
- Displays real-time feedback on the success or failure of function calls, with transaction details logged for user reference.

Project Timeline and Deadline

The timeline for the project was set to 1 month of backend work (development of apis) being the most important work ,1 week for the integration of a MySQL Database with proper schema , then the rest 2 weeks for the development of a user interface with functionalities within the next week .Then rest of the time for a design process .

3. Accomplishments and Achievements

A key accomplishment of this internship was the successful development of a fully integrated blockchain web application, utilizing a robust tech stack including Node.js, Express, React, SQL, and ethers.js. The project not only demonstrated a comprehensive understanding of blockchain technology and smart contract management but also showcased the ability to design and implement efficient APIs and a user-friendly interface, all backed by a reliable database system. This work provided a strong foundation for future blockchain projects within the company.

One of the major challenges I faced during this project was the connection of the web app to the blockchain network keeping in mind to use the latest libraries like the new ether.js v6 and this forced me to use the documentation of these packages to achieve this ,it was very new learning and it taught me a lot about the use of documentation and how to effectively search and debug

4. Lessons Learned

Throughout my internship, I gained invaluable knowledge and developed essential skills that significantly enhanced my understanding of blockchain development, REST API integration, and database management. Immersing myself in the full development cycle, I built a strong foundation in both server-side and client-side technologies like Node.js, Express, React, and SQL. A major takeaway was not only learning about blockchain's underlying principles, including consensus mechanisms and smart contract execution but also understanding the practical reasons behind its adoption. My deep dive into blockchain technology, combined with hands-on experience in implementing REST APIs that interact with smart contracts, emphasized the importance of using proper database schemas to ensure efficient and secure data management. Additionally, I developed a solid grasp of cryptocurrencies, including their structures and key concepts like gas prices, wei, and gwei, which are integral to blockchain transactions.

This onsite internship also provided valuable insights into the dynamics of a professional workspace, especially in terms of production environments and team collaboration. Working closely with a team, I learned how to navigate the challenges of integrating cutting-edge technologies, effectively use documentation, and contribute to collaborative problem-solving. The experience of connecting the web app to the blockchain network and managing data with proper schemas in a production setting reinforced the importance of well-structured, maintainable code. It also highlighted the critical role of communication and teamwork in successful project execution. Moving forward, I plan to continue refining my skills in developing modular, scalable REST APIs and to deepen my understanding of blockchain technology, with a particular focus on its application in collaborative, production-level environments.

5. Conclusion

In conclusion, my internship was a transformative experience that allowed me to develop a wide range of technical skills and gain valuable insights into blockchain development, API integration, and database management. A key accomplishment was the successful creation of a blockchain web app that integrated Node.js, Express, React, SQL, and ethers.js, showcasing my ability to work with complex systems and implement effective solutions for real-world challenges. This project, which included the development of robust APIs, the use of proper database schemas, and the seamless connection to the blockchain network, was instrumental in enhancing my problem-solving skills and deepening my technical knowledge.

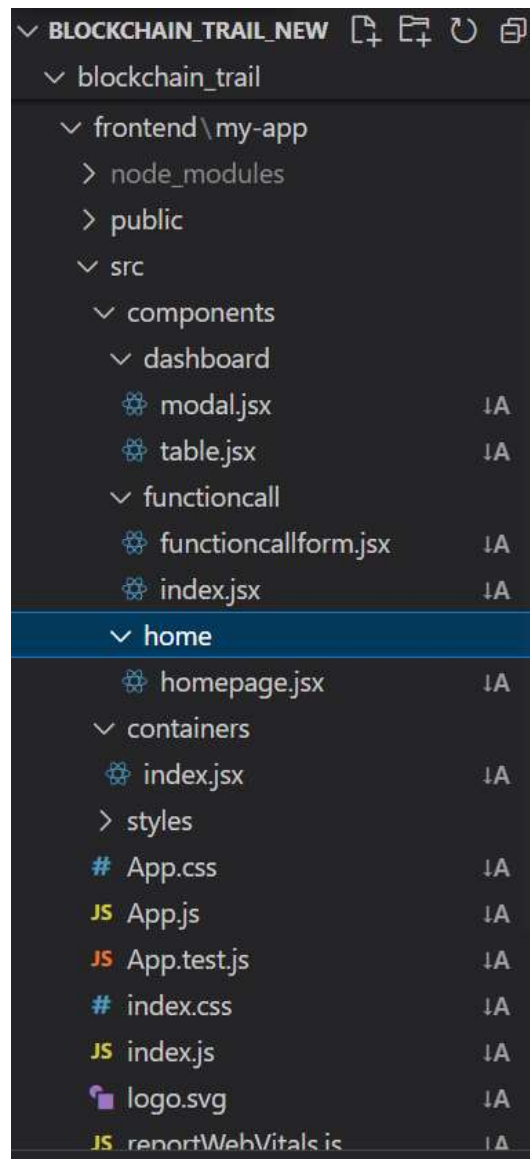
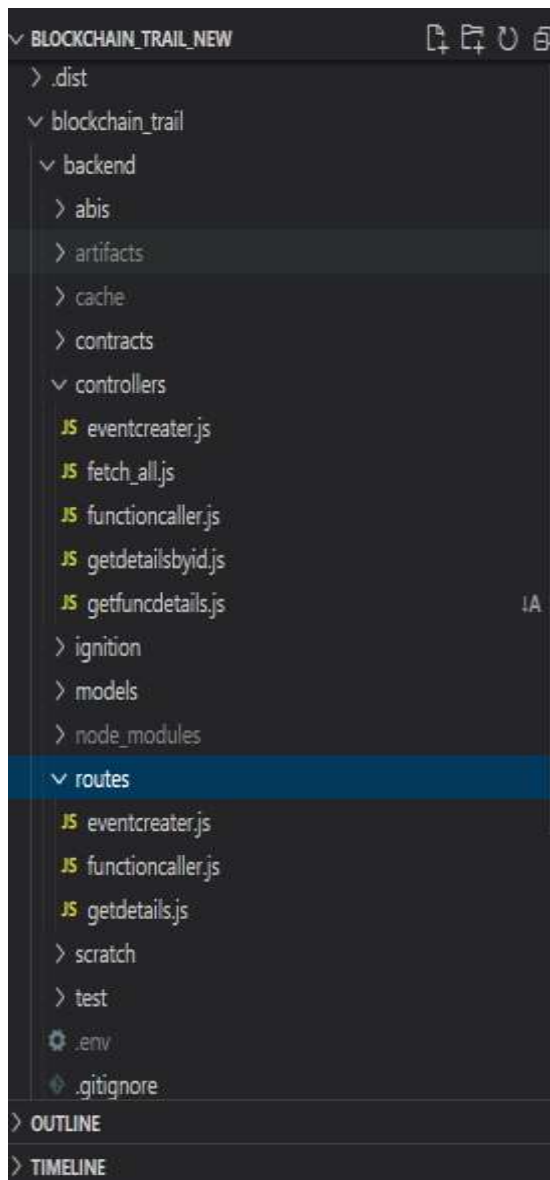
I am incredibly grateful to the company for providing this opportunity, and I would like to extend my sincere thanks to the mentors and colleagues who guided and supported me throughout the internship. Their insights, feedback, and encouragement were invaluable in helping me overcome obstacles and achieve success in my projects. I would especially like to thank Supreeth and the members of the blockchain team for their unwavering support and expertise, which were crucial to my professional growth and the successful completion of the project.

Based on my experiences, I would recommend that future internship programs continue to offer hands-on, project-based opportunities that allow interns to work on real-world challenges and develop practical skills. Providing more structured guidance on emerging technologies, such as blockchain and API development, would further enhance the learning experience. Additionally, fostering a collaborative environment where interns can work closely with experienced professionals and receive regular feedback will ensure that they gain the most from their time with the organization.

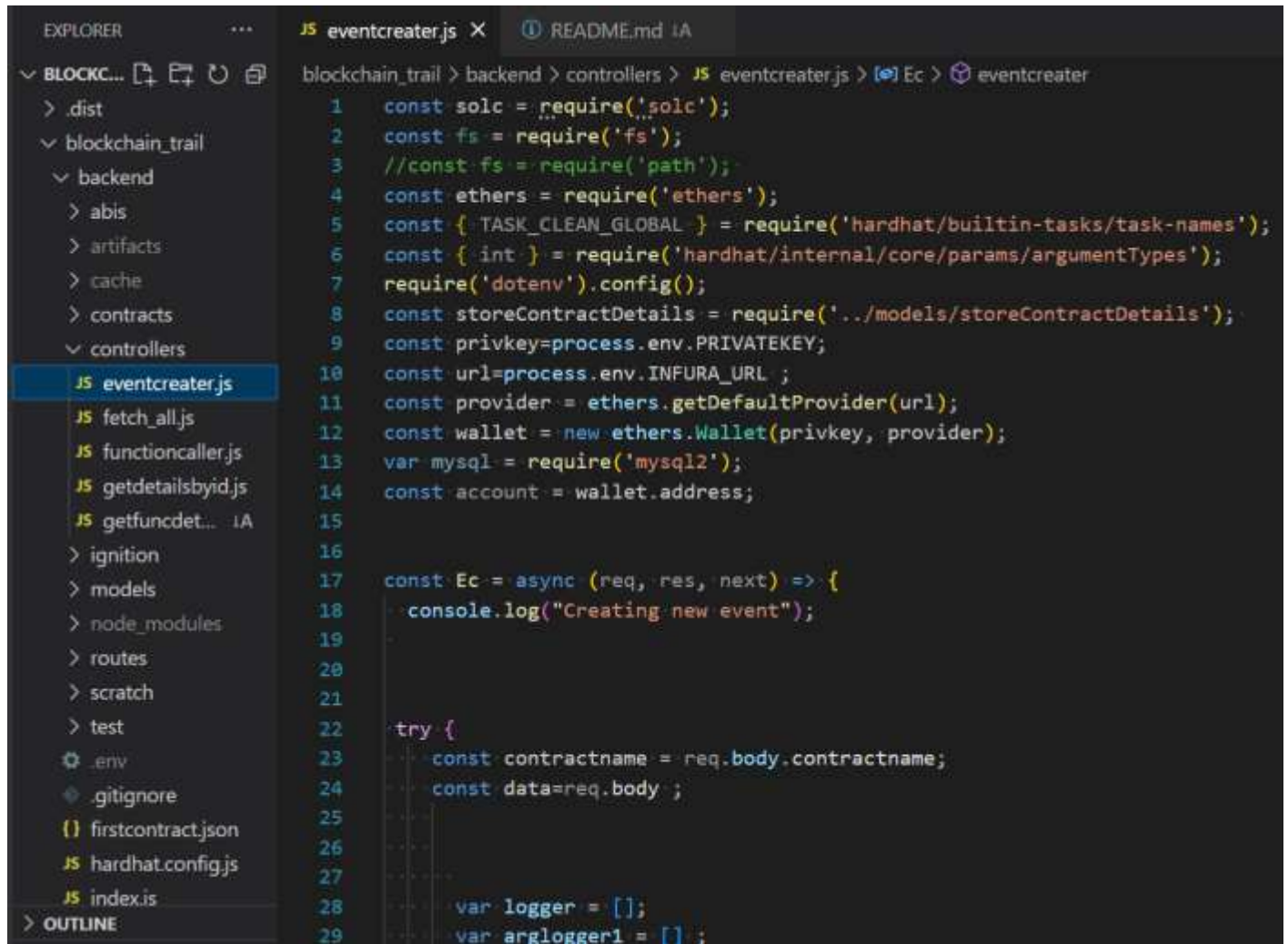
6. Appendices

GITHUB Repository: https://github.com/amogh545/blockchain_trail

Main File Tree:



Main api which creates the smart contract



```
1  const solc = require('solc');
2  const fs = require('fs');
3  //const fs = require('path');
4  const ethers = require('ethers');
5  const { TASK_CLEAN_GLOBAL } = require('hardhat/builtin-tasks/task-names');
6  const { int } = require('hardhat/internal/core/params/argumentTypes');
7  require('dotenv').config();
8  const storeContractDetails = require('../models/storeContractDetails');
9  const privateKey=process.env.PRIVATEKEY;
10 const url=process.env.INFURA_URL ;
11 const provider = ethers.getDefaultProvider(url);
12 const wallet = new ethers.Wallet(privateKey, provider);
13 var mysql = require('mysql2');
14 const account = wallet.address;
15
16
17 const Ec = async (req, res, next) => {
18   console.log("Creating new event");
19   -
20   -
21   -
22   try {
23     - const contractname = req.body.contractname;
24     - const data=req.body ;
25     -
26     -
27     -
28     var logger = [];
29     var arglogger1 = [] ;
```

1. Home page /Landing page



2. Dashboard – has the list of all contracts along with name, duration and navigation to the next page

ContractR

CREATE CONTRACT

Search Contracts

Contract Name	Duration	Action
amazon	4 days : 01 hours : 31 mins : 35 secs	Next
amazon	4 days : 01 hours : 33 mins : 52 secs	Next
amazon	4 days : 01 hours : 39 mins : 27 secs	Next
testfunc	12 days : 09 hours : 21 mins : 23 secs	Next
testfunc	13 days : 12 hours : 52 mins : 39 secs	Next
testfunc	21 days : 07 hours : 54 mins : 31 secs	Next
test5	24 days : 07 hours : 04 mins : 51 secs	Next
test4	24 days : 07 hours : 07 mins : 52 secs	Next
test3	24 days : 07 hours : 09 mins : 54 secs	Next

3. Create Contract form

Contract Name

Event Name

Input 1

Input 2
String

Input 1

Input 2
String

Event Name

Input 1

Input 2
String

+

×

−

+

−

+

+

−

+

SUBMIT

24 days : 07 hours : 09 mins : 28 secs

CREATE C



Search Contracts 🔍

1

Duration	Action
4 days : 01 hours : 40 mins : 10 secs	Next
4 days : 01 hours : 42 mins : 27 secs	Next
4 days : 01 hours : 48 mins : 02 secs	Next
12 days : 09 hours : 29 mins : 58 secs	Next
13 days : 13 hours : 01 mins : 14 secs	Next
21 days : 08 hours : 03 mins : 06 secs	Next

Call function page with event history

localhost:3001 coge
Function called successfully

Call Function:
addEventlaptopscount

_model (string)
dell

_count (uint256)
10

_ram (uint256)
8

_storageCap (uint256)
345

Calling...

Event History

#	model	count	ram	storageCap
1	lenovo	3	8	256
2	dell	10	16	1000
3	dell	10	8	345

_model (string)

_count (uint256)
0

_ram (uint256)
0

_storageCap (uint256)
0

Call Function

Event History

#	model	count	ram	storageCap
1	lenovo	3	8	256
2	dell	10	16	1000
3	dell	10	8	345

Shown along with the history ,this can also be viewed on polygonscan by anyone knowing the contract id thus achieving the goal of blockchain and maintains transparency